Measure sound speed

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Analyze sound travel measurements and their limitations.

SUCCESS CRITERIA - I can...

1. Define one-way versus round-trip paths.
2. Calculate speed from distance and time.
3. Use repeated measurements and discuss timing uncertainty.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “trial” and “echo” to explain your model or result. Sentence starters are optional.

trial: one run or repetition of a test

echo: a sound received after reflection

round trip: travel from start to a reflector and back

01 Bring back an earlier idea

Two wave displacements at one point are -3 mm and +7 mm.
Find the resultant and state whether they reinforce or partly cancel.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: v = d/Δt; v = 2 d/Δt for an echo.
Match the distance to one-way or round-trip travel. Use moderate sound levels and record actual data separately.

Worked reasoning: A sound echo returns 0.4 s after a pulse. At 340 m/s, round-trip distance is 136 m, so the reflector is 68 m away.

Your turn: Draw outgoing and returning paths and mark where the factor of 2 enters.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Look and explain

A pulse travels between two sensors separated by 1.70 m. Delay is .0050 s. Is this one-way or echo distance?
Find speed.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: ① ② ③

04 Build the solution

A toy pulse travels 12 m in 5 s.
Find a speed. If distance measurement is unchanged but time estimate is larger, would calculated speed rise or fall?

Word help: estimate = find an approximate value

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Use the idea

Plan a table for three trials. Include distance, elapsed time, method (one-way or echo), speed and one possible systematic error.

Sentence starter: The step I would change is … . The evidence is … .

Marzano check - circle: 1 2 3

06 Choose a method

Two equal-density spheres have radii r and 4 r. Supplied: area scales by 16, mass by 64.
Compare their area per mass.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

07 Exit check

Why does using d/t instead of 2 d/t for an echo give an incorrect speed?

Sentence starter: The step I would change is … . The evidence is … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Return to the worked example in problem 2. Two reflectors return echoes at 0.2 and 0.6 s.
Find both ranges and their separation along the same line.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3